RNGD Hotel Floor Plan Optimization: fp1_multi, Extended Program (3 Levels, 48 Keys)

RNGD design optimizer — 2026-10-10 22:59 — site fp1_multi, program hotel_program_extended.json

AI disclosure. The prose sections were written by anthropic/claude-sonnet-5.5 from the computed results, grounded with retrieved code passages. All tables and figures come directly from the data. 67 numbers in the AI text were checked against the data: 67 traced.

1 Executive summary

The optimizer placed all 62 of 62 rooms and all 48 requested keys across three levels on the fp1_multi site, with a final fitness of 0.9119. The best fitness in the first generation was 0.8972. Daylight, egress, ADA, and corner objectives all scored 1.0, and every exterior-facing check in the data came back at an exterior fraction of 1.0. All six code checks listed in the data pass, including the 50 ft dead-end limit in IBC-R1-027 (provided: 41 ft). The weakest objectives are efficiency (0.6763) and compactness (0.6412), and balance (0.816) is also moderate. The run was short (219 evaluations, 5.6 s) and did not use an LLM review, so the result is a first-pass layout that still needs professional code review.

Measure	Optimized design
Rooms placed	62 / 62
Keys placed	48 / 48
Placed area	22,831 SF of 22,576 SF
Corridor	936 SF per level (ratio 0.0971)
Rooms with a window (where needed)	1.0
Max exit travel / dead end	60.1 ft / 40.7 ft
Cores reached	2 / 2
ADA mobility rooms	5 placed, 2 required
Shared walls	948 ft
Fitness	0.9119

2 Approach

Rooms were placed on the fixed site grid with the systematic search method, using a population of 24 for 30 generations (seed 7). The run used 219 evaluations in 5.6 seconds. Fitness is a weighted sum of ten 0-1 objective scores. Placement carries the most weight (0.3), followed by daylight (0.14), egress (0.12), efficiency (0.1), ADA (0.08), core access (0.07), adjacency (0.07), balance (0.05), compactness (0.04), and corner (0.03). No LLM design review was used (use_llm is false, with 0 checkpoints), so the score reflects only the computed objectives. The best fitness rose from 0.8972 in the first generation to 0.9119 in the last.

3 Results

All 62 rooms and all 48 requested keys were placed. The placed area is 22831 sf against a program area of 22576 sf. Corridors take 936 sf, a corridor ratio of 0.0971. Maximum travel is 60.1 ft, the longest dead end is 40.7 ft, and both cores are reached. Daylight scores 1.0, with an exterior fraction of 1.0. Shared wall length is 948 ft. By level: L1 has 20 rooms, 12 keys, and 7820 sf, and also holds the public and back-of-house spaces (lobby/front desk, breakfast/lounge, fitness, luggage, public restrooms, BOH/office). L2 has 22 rooms, 19 keys, and 7827 sf. L3 has 20 rooms, 17 keys, and 7184 sf. The key mix is 7 King, 3 QQ, 1 King Suite, and 1 ADA King on L1. L2 has 10 King, 6 QQ, 1 King Suite, 1 ADA King, and 1 ADA Suite. L3 has 9 King, 5 QQ, 1 King Suite, 1 ADA King, and 1 ADA Suite. Efficiency (0.6763) and compactness (0.6412) are the lowest scores, and balance is 0.816. Core access is 0.8951 and adjacency is 0.9483. The data does not say what drives the lower efficiency and compactness scores.

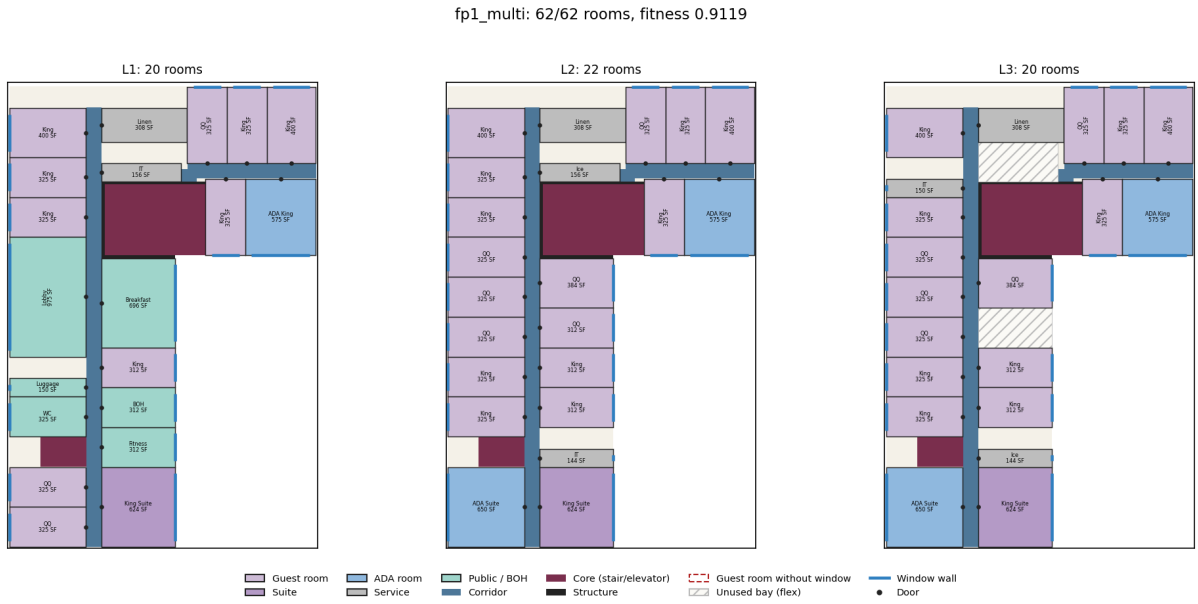


Figure 1: Optimized plans, all levels (corridor blue, cores maroon).

Level	Rooms	Keys	Area SF	Mix
L1	20	12	7,820	ADA King 1, BOH / Office 1, Breakfast / Lounge 1, Fitness 1, Housekeeping / Linen 1, IT / Elec 1, King 7, King Suite 1, Lobby / Front Desk 1, Luggage 1, Public Restrooms 1, QQ 3
L2	22	19	7,827	ADA King 1, ADA Suite 1, Housekeeping / Linen 1, IT / Elec 1, Ice / Vending 1, King 10, King Suite 1, QQ 6
L3	20	17	7,184	ADA King 1, ADA Suite 1, Housekeeping / Linen 1, IT / Elec 1, Ice / Vending 1, King 9, King Suite 1, QQ 5

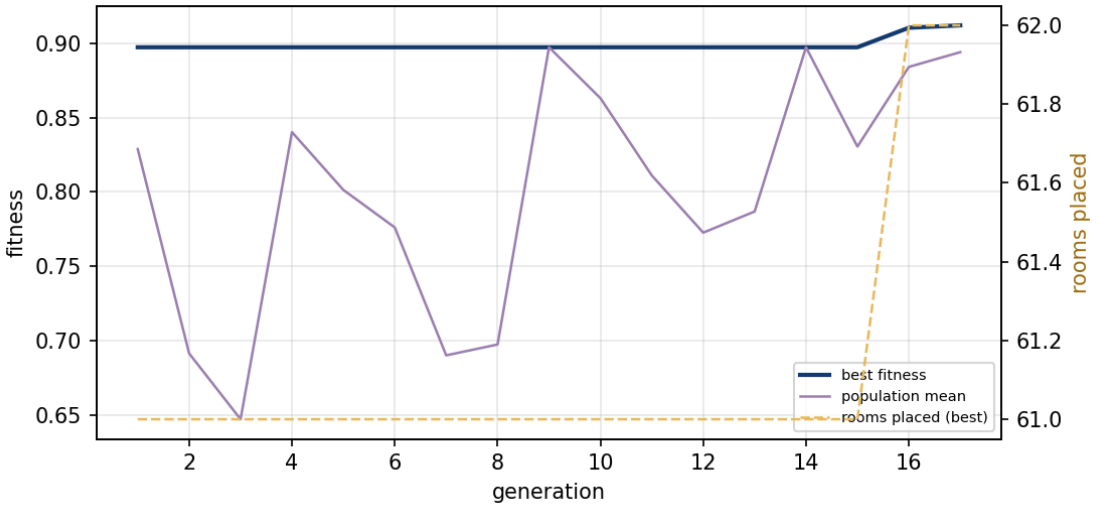


Figure 2: Convergence: best and mean fitness per generation; dotted lines mark LLM reviews.

4 Code and accessibility

All six checks in the data pass. For egress, IBC-R1-015 requires at least 2 exits and 2 are reached. IBC-R1-018 (exit separation) requires 60 ft and the layout provides 81 ft. IBC-R1-023 (exit access travel) allows up to 250 ft and the layout provides 60 ft; the zoning excerpt's stricter 200 ft limit to a stair enclosure would also be met. IBC-R1-025 requires a 44 in corridor and 60 in is provided. IBC-R1-027, the dead-end limit for a fully sprinklered R-1 building, is 50 ft, and the layout provides 41 ft. That rule is marked critical, starter-verified, and requires professional review, so the pass is a screening result and not a code approval. For accessibility, ADA-224.2 is checked at a minimum of 2 mobility rooms for 48 keys, and 5 are provided (one on L1, two each on L2 and L3). They are split between ADA King and ADA Suite types, which is consistent with the dispersion intent of ADA 224.5. The retrieved excerpts do not include the Table 224.2 values, so the required count of 2 comes from the check output and not from the excerpt. The communication-features table (224.4) shown starts at 151 rooms and was not evaluated for this 48-key hotel. IBC-R1-024 requires a 0.5-hour corridor rating, but no check for it appears in the data.

Rule	Check	Required	Provided	Status
IBC-R1-015	Exits per story (cores reached)	≥ 2	2	PASS
IBC-R1-018	Exit separation (core to core)	$\geq 1/3$ diagonal = 60 ft	81 ft	PASS
IBC-R1-023	Exit access travel	≤ 250 ft	60 ft	PASS
IBC-R1-025	Corridor width	≥ 44 in	60 in	PASS
IBC-R1-027	Dead-end corridor	≤ 50 ft	41 ft	PASS
ADA-224.2	ADA mobility rooms	≥ 2 for 48 keys	5	PASS

5 Comparison with the reference design

Note: the reference was made for the QA program (rooms_hotel.json) and this design for hotel_program_extended.json; room counts are therefore not directly comparable, while the code checks and efficiency measures are.

The reference design (floorplan1.json, 500 generations) reached a fitness of 0.8798 and placed 61 of 72 rooms and 58 keys, with no failed checks. The optimized design reached 0.9119 in 30 generations and placed 62 of 62 rooms and 48 of 48 keys. Its corridor ratio is 0.0971, against 0.1515 for the reference. Its exterior fraction is 1.0, against 0.8793. Its dead end is 40.7 ft, against 47.7 ft. It performs worse on two measures: maximum travel is 60.1 ft against 49.2 ft, and shared wall is 948 ft against 982 ft. The two designs appear to use different programs (62 versus 72 rooms and 48 versus 58 keys), so the fitness values are not a like-for-like comparison. The reference also did not place all of its rooms.

Measure	Reference (floorplan1.json, 500 gens)	Optimized
Rooms placed	61	62
Keys placed	58	48
Fitness (same evaluator)	0.8798	0.9119
Corridor ratio	0.1515	0.0971
Window fraction	0.8793	1.0
Max travel (ft)	49.2	60.1
Dead end (ft)	47.7	40.7
Cores reached	2	2
Shared walls (ft)	982	948
Failed code checks	none	none

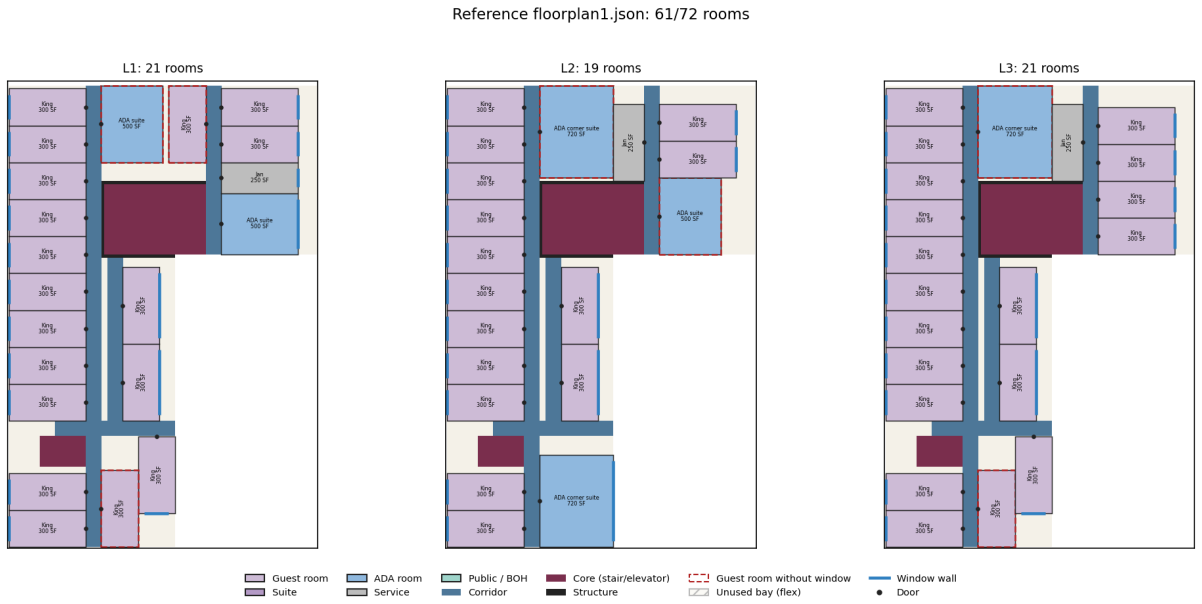


Figure 3: Reference design floorplan1.json

6 Objectives and weights

Objective	Meaning	Score	Final weight
placement	share of the program placed (priority weighted)	0.929	0.300
daylight	rooms that need a window have an exterior wall	1.000	0.140
efficiency	corridor area kept low	0.676	0.100
egress	exit travel and dead-end corridor within IBC limits	1.000	0.120
ada	ADA mobility rooms: count (Table 224.2) and spread over floors	1.000	0.080
core_access	service and ADA rooms open close to a core/elevator	0.895	0.070
adjacency	required room adjacencies met	0.948	0.070
corner	suites on corners	1.000	0.030
balance	keys spread evenly over floors	0.816	0.050
compactness	rooms share walls (compact, prefab-friendly)	0.641	0.040

7 LLM design reviews (inside the fitness function)

No reviews in this run.

8 Recommendations

- Have a licensed professional review the egress results, especially the critical dead-end rule IBC-R1-027 (41 ft against a 50 ft limit), before the layout is used for permitting.
- Add checks for corridor fire-resistance rating (IBC-R1-024) and for ADA communication-feature rooms, which the current output does not cover.
- Run a longer search (more generations or more seeds) and compare the results, since this run used only 30 generations and 219 evaluations.

- Investigate the low efficiency (0.6763) and compactness (0.6412) scores to see whether the objective weights or the floor geometry cause them.
- Run the optimization with the LLM design reviewer turned on to get a qualitative check on circulation and room adjacency.
- Confirm with the developer whether the 48-key program is the intended target, because the reference design had 58 keys.

9 Limitations

- Only one search run (seed 7) is reported, so the variability between runs is unknown.
- No LLM review was used, so the result reflects computed objectives only.
- Code rules are starter-verified, and the retrieved excerpts mark professional review as required. The pass results are not a compliance determination.
- The reference design appears to use a different program, so the comparison is indicative only.
- Some requirements are not modeled or not shown in the data, such as corridor ratings, ADA communication features, and accessible-room details such as bathroom type.

Appendix: fact check of the AI text

Number in text	Result
0.03	traced to data
0.04	traced to data
0.05	traced to data
0.07	traced to data
0.08	traced to data
0.0971	traced to data
0.1	traced to data
0.12	traced to data
0.14	traced to data
0.1515	traced to data
0.3	traced to data
0.5	traced to data
0.6412	traced to data
0.6763	traced to data
0.816	traced to data
0.8793	traced to data
0.8798	traced to data
0.8951	traced to data
0.8972	traced to data
0.9119	traced to data
0.9483	traced to data
1.0	traced to data
12	traced to data
15	traced to data
151	traced to data
17	traced to data
18	traced to data
19	traced to data
20	traced to data
200	traced to data
219	traced to data

Number in text	Result
22	traced to data
224.2	traced to data
224.4	traced to data
224.5	traced to data
22576	traced to data
22831	traced to data
23	traced to data
24	traced to data
24.2	traced to data
25	traced to data
250	traced to data
27	traced to data
27,	traced to data
30	traced to data
40.7	traced to data
41	traced to data
44	traced to data
47.7	traced to data
48	traced to data
49.2	traced to data
5.6	traced to data
50	traced to data
500	traced to data
58	traced to data
60	traced to data
60.1	traced to data
61	traced to data
62	traced to data
7184	traced to data
72	traced to data
7820	traced to data
7827	traced to data
81	traced to data
936	traced to data
948	traced to data
982	traced to data